Rehmat Ali

(Embedded AI Engineer)

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Summary

I'm a Master's student at Polytech Nantes, working at the intersection of Artificial Intelligence and Embedded Systems. While many see AI as a cloud-first technology, I'm fascinated by the challenge of making it run efficiently on the devices we use every day. My work focuses on Edge AI: developing and optimizing models for real-time performance on low-power hardware. I'm driven to build the next generation of smart, autonomous, and efficient embedded applications.

This specialization builds on my 3.5-year foundation as a software developer, where I engineered robust ERP systems for international clients. I am now focused on leveraging this combined expertise in reliable software and embedded AI to build the next generation of intelligent, autonomous devices.

Education

Master's In Embedded Technologies and Artificial Intelligence Nantes Université, Nantes, France — Nov. 2025 to Aug. 2026

Description: The Master's in Embedded Technologies and Artificial Intelligence focuses on integrating advanced AI algorithms into embedded systems. It combines hardware design, software engineering, and machine learning to develop intelligent, efficient, and autonomous devices capable of real-time decision-making on low-power hardware.

Bachelor of Science (Software Engineering) SZABIST, Islamabad — Sept. 2017 to July 2021

GPA: 3.16/4.0
Majors: Data Science, Digital Image Processing, Artificial Intelligence, Software Engineering, Object Oriented Programming, Software Architecture & Design and Visual Programming,

Experience

ERPHUB, ERP Solution Devleoper — Cleveland, USA

Onsite: Islamabad

Jan 2023 to Aug 2025

2.8 years

- Designing custom applications and automating workflows using Zoho Creator and Deluge scripting
- Developing and optimizing processes across Zoho CRM, Books, People, and Desk
- Building interactive dashboards and data visualizations in Zoho Analytics
- Integrating Zoho apps with external platforms using REST APIs, webhooks, and custom functions
- Implementing AI-based solutions within the Zoho environment such as:
  * Predictive lead scoring
  * Smart automation flows
  * Chatbots for customer support
  * AI-powered analytics for business forecasting
- Supporting smart, scalable, and data-driven applications that align with evolving business goals.

Link: Experience Letter

FairchanceforCRM Zoho Developer

Lahore, Pakistan

Onsite

Aug 2021 to Dec 2022

1.5 years

- Held regular meetings with clients to gather and analyze business requirements for Zoho-based solutions.
- Planned and implemented customized workflows across various Zoho applications to streamline business processes.
- Designed and developed custom workflows and functions to automate repetitive tasks and enhance CRM efficiency.
- Developed and trained a SalesIQ chatbot to handle multiple customer scenarios, trigger actions, and provide contextual responses, improving the overall digital support experience.

Link: Experience Letter

Projects

Deep Learning Based Framework for Colorectal Polyp Classification

Developed an AI-powered diagnostic system for early detection of colorectal cancer by classifying colonoscopy images of intestinal polyps using Convolutional Neural Networks (CNN). The system distinguishes between adenomatous and hyperplastic polyps with 94% accuracy, enabling faster and more reliable diagnoses. Built as a webbased platform with modules for doctors and patients, the tool allows uploading colonoscopy images and generates classification reports within minutes. The project followed the Spiral Model of SDLC for iterative development and risk management.
Link: Documentation

Voice-Controlled Home Automation using Arduino

This project is a home automation system designed to control electronic devices using voice commands. The core of the system is an Arduino board programmed in C++, which acts as the central controller. The system is operated via a mobile phone application that captures voice commands, such as "turn on" and "turn off." These commands are transmitted wirelessly to the Arduino board using a Bluetooth module. Upon receiving a command, the Arduino processes the instruction and controls the power state of LEDs connected to it, demonstrating a proof-of-concept for managing household lights and appliances. This project showcases skills in C++ programming, hardware integration with Arduino, and wireless communication via Bluetooth.

Technologies

Languages: Python, PHP, SQL, C, C++, Javascript, Deluge

Front End Development Techniques: HTML, CSS, Javascript, Bootstrap, JQuery

Frameworks: Tensorflow, Jupyter Notebook, Google Colab, Overleaf

Languages

English (Fluent) Urdu (Native) French (Basic)

REFERENCES

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